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Innovation education and entrepreneurial intentions among postgraduate students: The role of innovation competence and gender

Faisal Iddris¹, Courage Simon Kofi Dogbe^{1*} and Emmanuel Mensah Kparl¹

Abstract: The study examined the effect of innovation education on entrepreneurial intentions among postgraduate students and the role of innovation competence and gender. The study was a survey, with data collected using a structured questionnaire. The study used Master's students of Akenten-Appiah Menka University of Skills Training and Entrepreneurial Development (AAMUSTED), Ghana. The sample includes 370 Master's students from 10 faculties of the university. Data were analyzed using Structural Equation Modelling (SEM). The study found that innovation education has significant positive influence on entrepreneurial intention. Innovation competence was found to mediate the relationship between innovation education and entrepreneurial intention. Finally, gender moderated the relationship between innovation education and innovation competence and also moderated the relationship between innovation competence and entrepreneurial intention. Assessing the moderating role of gender between innovation education and innovation competence, as well as, innovation competence and entrepreneurial intention, represents novel contributions of this study.

Subjects: Entrepreneurship and Small Business Management; Educational Research; Higher Education

Keywords: Innovation education; innovation competence; entrepreneurial intention; gender; postgraduate student

1. Introduction

Entrepreneurship has become one of the major ways of sustaining economic development ever. Entrepreneurship has turned out to be a key contributor to economic development by sustaining development and improving economic growth in the 21st century (Pizzi et al., 2020; Veleva, 2021). Entrepreneurship in these recent times has become the leading drive for sustainable development, by contributing greatly to economic growth and reducing unemployment in the global economy (Urbano et al., 2020). Governments therefore seek diverse ways in enhancing the entrepreneurial potentials of their students. To create more awareness about entrepreneurship, many countries of late are focusing and investing more in entrepreneurial and innovation education (Ismail et al., 2009). Innovation education is the process that enlightens potential entrepreneurs with the skills and knowledge that provide them with the opportunities to identify appropriate business areas and to allocate the needed resources for starting a business venture in the long run (Lee & Yuan, 2018). Tertiary and other educational institutes are focusing more on theoretical work rather than practical support and developing competencies (Usman et al., 2020), which should not be the ideal situation.

A number of studies have suggested that education influences a person's level of entrepreneurship (Lackéus, 2020). The kind of knowledge and skills that individuals gain during the tertiary level of their education contribute significantly to their entrepreneurship intention (Aboobaker & Renjini, 2020). Studies such as Li et al. (2020) predicted entrepreneurship behavior among students, using entrepreneurial passion, alertness, self-efficacy, and proactive personality. Boahemaah et al. (2020) also identified that entrepreneurship education had a direct positive effect on entrepreneurial intentions. Yet, the study by Bamfo et al. (2017) pointed out that, entrepreneurship education had no significant effect on entrepreneurial intentions among students. Others have the belief that people are born with entrepreneurial tendencies and education cannot add to people's perception about entrepreneurship (Jones & English, 2004). Laukkanen (2000) argued that formal education demotivates people from starting a new venture. Aside from these inconsistencies, not much attention has been paid specifically to the influence of innovation education on entrepreneurial intention, making this present study even more relevant.

The main focus of this study is to examine the effect of innovation education on entrepreneurial intentions among postgraduate students and the role of innovation competence and gender. Innovation education explains how students are being trained to acquire the needed skills to become innovators to start their own business venture (Lee & Yuan, 2018). Innovation education is the training that students and other individuals go through to gain innovative ideas in the society (Córdoba-Pachón et al., 2021). Innovation education is where people are groomed in terms of ideas, knowledge, and skills on how people can use their abilities to enter into private business (Rosa, 2020). Innovation education inspires students to research, discover, and use all the tools to explore something new. Innovation education involves diverse ways of identifying problems in the business field and providing strategies for solving them. It also improves entrepreneurial intention because it compels students to use a higher level of thinking to solve complex problems on how to create new business (Usman et al., 2020). For graduate students to have entrepreneurial intentions, they need to adopt the skills they gain during innovation studies (Usman et al., 2020). Innovation education is thus a necessary tool that helps to enhance students' entrepreneurial intention by improving the competencies and capabilities. Innovation education builds complex skills in graduate students; this means that it encourages them to use complex tools, ideas, and skills to start and complete a project (Biasi et al., 2021).

Innovation competence does not take place in isolation; it mostly depends on innovation education which results in empowering students' entrepreneurial intention. Innovation education builds multiple skills in students to gain adequate knowledge and shapes them on how to build entrepreneurship projects (Aburai & Takeyasu, 2020). Students are able to establish business ventures successfully through the knowledge gained from innovation education. This also enhances students' performance after establishing business. Innovation education enables students to demonstrate proper behavior in maintaining business ventures. Students who are able to adequately adopt their entrepreneurial intention are those who gain proper skills from innovation education. Innovation education helps students to be competent by ensuring how to find innovative objects to start and maintain a new business. Innovation helps students to build their innovation competence which results in creating new business (Sabarguna, 2017). The study will contribute to existing knowledge by establishing how students' innovation competence will mediate between innovation education and entrepreneurial intention among postgraduate students.

Moreover, although innovation education seems to have a direct influence on innovation competence, there could be some potential variables such as gender (being male or female) intervening in the relationship (Adeyemo et al., 2021). Innovation education enhances students' competencies when it comes to setting up a business. Educating students about innovation creates an avenue for students to gain upper hand in generating business ideas. Male students are motivated well enough after receiving innovation education which gives them the morale to start a business than female. Male students who receive innovation education possess a high level of innovation competence (Gurel et al., 2021). Thus, the next input of the study is the potential

moderating role of gender in the relationship between innovation education and innovation competence among postgraduate students.

Postgraduate students demonstrate innovation competence which affects their intention and readiness to become an entrepreneur to start a new business. Postgraduate students in recent times are lacking adequate and qualified skills and the competencies that will drive them to enter into new business on their own (Allan, 2021). Most graduates after school do not think about creating new business venture but rather, they always focus their attention on doing white-collar jobs which in our world today has become very difficult (Ajamobe, 2021). This is as a result of students not having requisite qualities and the ability to think about creating something independently. The role of innovation has become more important in our world today when it comes to business creation. Innovation competence gives students the chance to enter into entrepreneurship. The skills and knowledge students receive from innovation education enhances their ability to adopt their entrepreneurial intention. Innovation competence improves students' intention to become an entrepreneur after receiving training on how be creative and to think innovatively. Male students who receive innovation education are more likely to gain greater innovation competence than female students. Males are found to have more knowledge and adequate skills in innovation than females after going through innovation education (Gurel et al., 2021). Male students after going through the required training are able to deliver at the start-up of creating a business venture than female students. Men are known to be equipped with more qualities that are relevant for starting a new business venture. The third contribution of this study focused on the moderating role of gender in the relationship between innovation competence and entrepreneurial intention of postgraduate students. Figure 1 presents the conceptual framework of the study.

2. Literature review

2.1. Innovation education and entrepreneurial intention

Education deals with how people acquire skills, knowledge, and new ideas, generated either through formal or informal settings which affect the way an individual thinks, feels, and acts at a given point (Dunne, 2021). Educating students on how to be business-oriented and think innovatively is one of the ways that helps students to become entrepreneurial minded or have the desire to start a new business (Borgen & Dalgaard, 2021). Innovation education is the process of training students and other individuals to gain new innovative skills, knowledge, and the ability to develop ideas to solve societal problems (Córdoba-Pachón et al., 2021). To be employed after school is one of the main motives of every student. However, through education students are able to become creative entrepreneurs. Education enlightens students' understanding and the potential benefits of becoming entrepreneurs. Postgraduate students would stand in a better position to start their own private business through the training they gain from school. That is, postgraduates are more likely to be industrious and entrepreneurial minded and have good interest and the intention to start a business right after graduation and this can be done after going through innovation education (Saji & Nair, 2018). According to Usman et al. (2020), innovation education has a direct positive relationship with entrepreneurial intention. The education people gain on innovation gives them the advantage to be exposed to entrepreneurship. Innovation education significantly contributes to how students successfully establish new venture after acquiring the requisite knowledge and skills in entrepreneurship. According to Laguía et al. (2019), postgraduate students with good intention to set up a new business and deciding to become entrepreneurs are students who have in-depth education on innovation. In a situation where innovation education and training are regarded as one of the major ways to improving students' intention on entrepreneurship, setting up a new business never becomes a problem (Wilson et al., 2007). Students who get the opportunity to study and also have training on innovation education stand a better chance to start a new venture (Laguía et al., 2019). Innovation education inspires and motivates postgraduate students to become entrepreneurs (Souitaris et al., 2007).

E. M. Rogers developed one of the oldest social science theories called Diffusion of Innovation (DOI) Theory in the 1960s. This theory talks about how individuals are able to adopt new ideas skills and knowledge on technologies that emerge into the system (Miller, 2015). Individuals must perceive the ideas, behavior, technology, and product as new innovative ways that come through innovative education (Miller, 2015). Innovation education gives postgraduate students the ability to create new products when it comes to setting up new business and delivering proper service (Biasi et al., 2021). Educating postgraduate students on innovation helps them to be more focused and relaxed when entering into entrepreneurship because it gives students the hope for performing tasks easily in business setup and operations and also provides the opportunity for postgraduate students to follow the new trend of doing business (Zhuge et al., 2021). Innovation education gives postgraduate students the skills, knowledge, and ideas on how to handle and work with current tools and instruments and how to use them to create something new in business operations (Vincent-Lancrin et al., 2017). Postgraduate students gain more interest and have proper intention for entrepreneurship through these innovation education factors. Grounded in human capital theory, Bae and Patterson (2014) found in their study that education has a direct effect on students' entrepreneurial intention. Therefore, we hypothesize that

H1: Innovation education has a direct positive effect on entrepreneurial intention among postgraduate students.

2.2. Mediation role of innovation competence

Innovation competence can be defined as the effort to increase performance by promoting new things such as ideas, knowledge, and skills to achieve quality results (Ovbiagbonhia et al., 2019). Mulder (2014) has grouped competencies into three perspectives: behavioral functionalism, integrated occupationalism, and situated professionalism. This study will follow Mulder (2014) explanation of competence as situated professionalism, as it is stated that competence is when in specific-context professionals like postgraduate students gain new skills, knowledge, methods, and standards for doing something with ease. Competence is the assessment of an act in a specific activity (Kyndt & Baert, 2015). Innovation education builds multiple skills in students to gain adequate knowledge and gives them the ability on how to start business. It enables postgraduate students to demonstrate proper behavior in establishing and maintaining business ventures. Students after gaining competence in innovation are, thus, able to generate proper intentions in entrepreneurship. Innovation competence creates room for students to start a business venture. The ideas that postgraduate students generate to start a new business are as a result of having adequate knowledge and ability gained through innovation education. The theory of competence-based theory (Sanchez, 2004) explains how individuals can use strategic and dynamic ways to accomplish successful task from insight gain. This theory explains how to holistically put in every effort to achieve a set target. According to Keinänen and Kairisto-Mertanen (2019), students who learn more on innovation gain innovation competency and also have the intention to become entrepreneurs. Innovation education assists postgraduate students to be competent when it comes to business venture creation. Studies have shown that innovation education has direct effect on entrepreneurship intention (Usman et al., 2020); however, the effect could also be realized through the intervention of innovation competence among postgraduate students. Based on the discussion we, therefore, hypothesize that

H2: Innovation competence mediates the relationship between innovation education and entrepreneurial intentions among postgraduate students.

2.3. Moderating role of gender in the relationship between innovation education and innovation competence

Researchers have researched on what differentiates men and women regarding several aspects of competencies. For instance, women are seen to have more innovation competencies in starting a business venture after going through innovation education (Wu et al., 2021). Likewise, men are perceived to be equipped with adequate skills and abilities when it comes to starting a business (Ferrerias-Garcia et al., 2020). These differences between males and females may generate a lot of different views among genders. When investigating the relationship between innovation education and innovation competence, another individual variable that affects these variables is gender (Gurel et al., 2021). Ferreras-Garcia et al. (2020) stated that men have more innovation competencies than women do when it comes to starting a business venture. Empirical studies have shown that males are found to be more innovative competent after going through innovation education and training (Wu et al., 2021). This signifies men's ability to demonstrate proper skills and have in-depth knowledge in business creation than women. Men are more competent when it comes to setting up a business because they engage themselves more in innovation education (Ferrerias-Garcia et al., 2020). Postgraduate male students after engaging in innovation education are highly above postgraduate female students when it comes to innovation competence. Men prove to be more focused and ever ready to learn new things and are always motivated to push further into business establishment (Barron et al., 2020). We, therefore, hypothesize based on the discussion that

H3: Gender moderates the relationship between innovation education and innovation competence, such that the effect of innovation education on innovation competence is higher among male postgraduate students.

2.4 Moderating role of gender in the relationship between innovation competence and entrepreneurial intentions

Several scholars have studied between genders that are more likely to grow entrepreneurial intention. Fayolle et al. (2015) indicated that postgraduate male students have higher intention on entrepreneurship due to their ability and positive attitude towards new business creation. Comparing males to females, males with adequate skills and ability in innovation are more likely to enter into venture creation than female students. For example, Schwarz et al. (2009) found men are always motivated to set up a business that women do. The theory of planned behavior (Ajzen, 1985) is a psychological theory that links beliefs to behavior. The theory maintains that three core components, namely, attitude, subjective norms, and perceived behavioral control, together shape an individual's behavioral intentions. Men are believed to enter into entrepreneurship because they know and understand their capabilities and abilities in innovation than women. According to Joensuu et al. (2013), men who are innovatively competent are more likely to become entrepreneurs than women. Empirically, postgraduate male students with innovation competence have a significantly higher intention towards entrepreneurship than female postgraduate students (Schwarz et al., 2009; Joensuu et al., 2013). Females tend to have less entrepreneurial intention; this shows that female postgraduate students do not express higher entrepreneurial intention than their males (Kurczewska & Bialek, 2014). We, therefore, hypothesize that

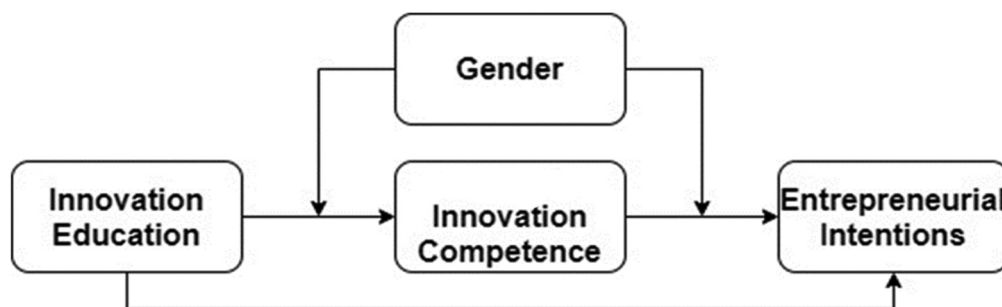
H4: Gender moderates the relationship between innovation competence and entrepreneurial intentions, such that the effect of innovation competence on entrepreneurial intentions is higher among male postgraduate students.

3. Research methodology

3.1. Research design

In terms of time design, the study was a cross-sectional research, by gathering data from a particular point in time (Saunders et al., 2015). In terms of design approach, the study was a quantitative research, by gathering data using structured questionnaire. In terms of design

Figure 1. Conceptual framework.



strategy, the study was an explanatory study, by conducting a cause-and-effect analysis using Structural Equation Modelling technique.

3.2. Collecting data

The study focused on Master's students of Akenten-Appiah Menka University of Skills Training and Entrepreneurial Development (AAMUSTED). The university has 10 faculties, which were Faculty of Business Education, Faculty of Technical Education, Faculty of Applied Sciences and Mathematics Education, Faculty of Technical Education, Faculty of Vocational Education, Faculty of Education and Communication Sciences, Faculty of Agriculture Science Education, Faculty of Education and General Studies, Faculty of Science Education, and Faculty of Environment and Health Education. The study sampled Master's students from all these faculties, to assess their behavioral intention to become entrepreneurs after school. This study is important, as a core mandate of the university, as could be seen from its name, is to train entrepreneurially minded graduates. Using a quota sampling technique, 370 students were sampled from the 10 faculties of the university, based on the student population of each faculty.

The questionnaires were administered by three researchers and seven Teaching and Research Assistants, who were all affiliated to AAMUSTED. Each member was responsible for data collection from one faculty. The study made use of printed questionnaires, as the respondents were easily assessed in person. Questionnaires were administered during class hours, after permissions were sought from the respective lecturers in charge of the various classes at the period of data collection. The data collection spans a period of 7 continuous days; this was to ensure that the weekend students were also captured in the study.

Structured questionnaire was used as the data collection instrument, and it had four sections. Section A presented the demographic characteristics. Under Section A, demographics such as faculty, level/year, age, and gender of students were assessed. Section B presented the measurement items under innovation education, which were to be answered as Yes or No. There were five measurement items under innovation education, which were adapted from Lee and Yuan (2018), Thorsteinsson (2014), and Maritz et al. (2014). Section C addressed individual student's innovation competence, and measurement items were responded to on a Likert scale of 1-strongly disagree to 5-strongly agree. The measurement items under innovation competence were adapted from Watts et al. (2013). The final section of the questionnaire addressed entrepreneurial intention among the postgraduate students, and statements were responded to on a Likert scale of 1-strongly disagree to 5-strongly agree. The measurement items under entrepreneurial intention were adapted from Narmaditya and Wibowo (2021).

(Table 1) presents the profile of the respondents to this study. Results presented demonstrated that the Faculty of Business Education had the highest representation in this study (21.35%), while the Faculty of Agriculture Science Education had the least representation of 7.03%. Final year students also dominated the study, representing 55.14%. Male respondents constituted 58.65%,

Table 1. Profile of respondents

Profile of Respondents	Frequencies (N)	Percentages (%)
<i>Faculty</i>	370	100%
Faculty of Business Education	79	21.35
Faculty of Applied Sciences and Mathematics Education	48	12.97
Faculty of Science Education	42	11.35
Faculty of Education and General Studies	41	11.08
Faculty of Education and Communication Sciences	39	10.54
Faculty of Environment and Health Education	37	10.00
Faculty of Technical Education	31	8.38
Faculty of Vocational Education	27	7.30
Faculty of Agriculture Science Education	26	7.03
Level	370	100%
1 st Year	166	44.86
2 nd Year	204	55.14
Gender	370	100%
Male	217	58.65
Female	153	41.35
Age	370	100%
Less than 25 years	40	10.81
25–35 years	163	44.05
35–45 years	134	36.22
45–55 years	31	8.38
Above 55 years	2	0.54

representing the highest gender group. Finally, it could be realized that respondents aged 25–35 years were 44.05%, while respondents aged 35–45 years were also 36.22%.

3.3 Choosing the proper model for analysis

We first run an Exploratory Factor Analysis (EFA) in SPSS (v.23), to assess if the measurement items properly loaded onto their corresponding latent variables. There were three latent variables in this study, which were innovation education, innovation competence, and entrepreneurial intention among the postgraduate students. Based on the EFA, measurement items which loaded poorly (less than 0.5) or cross-loaded on different construct were deleted from further analysis. Based on this principle, two measurement items were deleted from innovation education: three from innovation competence and one from entrepreneurial intention. From the EFA results, the total variance extracted (TVE) was 65.21% which was higher than the minimum expected score of 50%. To measure the sample adequacy, Kaiser–Meyer–Olkin (KMO) was assessed, and the score was 0.729, which was larger than the minimum expected score of 0.6. Bartlett’s Test of Sphericity was expected to be statistically significant, to mean there was enough correlation among the measurement items to qualify for EFA. Results from this study indicated a significance score ($\chi^2 = 1447.45$; $p < 0.01$). To achieve positive definiteness in the dataset, the determinant of correlation should not be equal to zero (0), and this study had a determinant of 0.002.

Table 2. Confirmatory Factor Analysis (CFA)

Model Fitness: CMIN = 415.91; DF = 148; CMIN/DF = 2.81; GFI = 0.875; PClose = 0.105; TLI = 0.936; CFI = 0.955; RMSEA = 0.037; RMR = 0.052	Std. Factor Loadings
Innovation Education (INN_EDUC): CA = 0.834; CR = 0.819; AVE = 0.602	
I have had classroom education on the concept of business innovation (IE1)	0.734
I have been taught on the need to be innovative in business operations (IE2)	0.828
My knowledge on business innovation has been assessed through formal examination (IE3)	0.763
I have had classroom education on the concept of creativity in business operations	D*
As part of my course assessment, I have ever been tasked to development a new product for an organization	D*
Innovation Competence (IIN_COMP): CA = 0.826; CR = 0.848; AVE = 0.530	
<i>I am able to ...</i>	
offer ideas that are original in content (IC1)	0.599
critically evaluate the fundamentals of ideas/actions (IC2)	0.829
propose suggestions/alternatives appropriate to the demands of tasks (IC3)	0.757
use resources creatively (IC4)	0.778
act persistently until I see results (IC5)	0.654
foresee how events will develop	D*
take intelligent risks	D*
set personal targets for tasks	D*
Entrepreneurial Intentions (ENT_INTENT): CA = 0.891; CR = 0.897; AVE = 0.642	
I am ready to do anything to become an entrepreneur (EI1)	0.792
My goal is to be an entrepreneur (EI2)	0.951
I will make every effort to start and run my own company (EI3)	0.846
I was determined to make the company of the future (EI4)	0.789
I have seriously thought about starting a business (EI5)	0.582
I have robust intentions to start a business in the near future	D*

D* ~ Items deleted due to poor factor loadings

Confirmatory Factor Analysis (CFA) was run in Amos (v.23), using the retained variables from the EFA. Similar to the EFA, the standardized factor loading from the CFA was expected to be a minimum of 0.5, which was achieved in this study (Table 2). The minimum factor loading under innovation education was 0.734, that of innovation competence was 0.599, and that of entrepreneurial intention was 0.582, indicating that all measurement items significantly explained their latent variables. Using the retained variables, Cronbach's Alpha (CA) was calculated with SPSS (v.23), with a minimum expected alpha score of 0.7. This was achieved for all the three latent

Table 3. Discriminant validity

	Mean	Std. Dev.	1	2	3	4	5	6	7
Faculty (1)	-	-	-						
Level (2)	-	-	0.197*	-					
Age (3)	-	-	0.223**	0.158*	-				
Gender (4)	-	-	0.175	0.251**	0.161*	-			
INN_EDUC (5)	3.281	0.644	0.481**	0.211*	0.124	0.137	0.776		
INN_COMP (6)	3.842	0.856	0.219**	0.141*	0.242**	0.551**	0.613**	0.728	
ENT_INTENT (7)	4.263	0.930	0.208**	0.343**	0.131	0.486**	0.624**	0.503**	0.801

**Sig. at 1%; *Sig. at 5%

√AVEs are bold and underlined

variables (Table 1), indicating there was high internal consistency (reliability) among the measurement items. Convergent validity was assessed by using Average Variance Extracted (AVE) approach, whose minimum score was supposed to be 0.5 Fornell and Larcker (1981), which was achieved for all the latent constructs. Composite reliability (CR), on the other hand, was expected to be at least 0.7, which was also achieved for all the latent variables.

When conducting CFA, it was essential to assess the model fit indices, to determine if the dataset appropriately fits the model estimated. As part of model fit indices, CMIN/DF should be less than 3, PClose should be statistically insignificant (> than 0.05), TLI and CFI should be larger than 0.9, while RMSEA and SRMR should also be less than 0.08 (Hair et al., 2010). These were achieved for all the latent variables in this dataset.

In assessing the discriminant validity of the measurement items, studies such as Bamfo et al. (2018) among others, compared the squared-root of the AVEs (√AVEs) with the inter-correlation scores. To claim for discriminant validity, the least √AVE was expected to be larger than the highest correlation score. From (Table 3), the lowest √AVE was 0.728, while the highest correlation score was 0.613. Discriminant validity is therefore said to have been achieved. Furthermore, it was concluded that there was no multicollinearity in the dataset, as the highest correlation coefficient of 0.613 did not exceed 0.8.

Table 4. Direct, moderating, and mediation path estimates

Direct Paths	UnStd. Estimate	S.E.	C.R.
INN_EDUC → INN_COMP	0.581	0.149	3.899**
Educ_Gen → INN_COMP	0.261	0.066	3.955**
Gender → INN_COMP	0.229	0.059	3.881**
INN_EDUC → ENT_INTENT	0.492	0.109	4.514**
INN_COMP → ENT_INTENT	0.267	0.101	2.644**
Gender → ENT_INTENT	0.105	0.047	2.234*
Comp_Gen → ENT_INTENT	0.388	0.106	3.660**
Faculty → ENT_INTENT	0.293	0.124	2.363*
Level → ENT_INTENT	0.023	0.031	0.742
Age → ENT_INTENT	0.024	0.029	0.828
Indirect Effect		Lower BC	Upper BC
INN_EDUC → INN_COMP → ENT_INTENT	0.155	0.009	0.481

Bootstrap Bias-Corrected Confidence Interval at 95%

**Sig. at 1%; *Sig. at 5%

4. Results

The path analysis was estimated using the Structural Equation Modelling (SEM) run in Amos (v.23), with results presented in (Table 4) and (Figure 2). The estimation was based on 5000 Bootstrap samples, with Bias-Corrected Confidence Interval of 95%. The study controlled for three variables, which were faculty of the students, level or year, and age of the student. The results as presented indicated that faculty had a significant positive effect on entrepreneurial intention ($\beta = 0.293; p < 0.05$). Faculty was coded as 1-Faculty of Business Education and 0-Others. This implies that faculty of business education students were 29.3% more likely to have entrepreneurial intentions than postgraduate students from other faculties of the university. Although level and age of students had positive effects, these effects were not statistically significant ($p > 0.05$).

Regarding the main paths, the results pointed out that innovation education had a significant positive effect on entrepreneurial intention among postgraduate students ($\beta = 0.492; p < 0.01$). Postgraduate students who have had formal education on innovation were 49.2% more likely to have entrepreneurial intention. This is because innovation forms a key component of

Figure 2. Diagrammatic presentation of SEM.

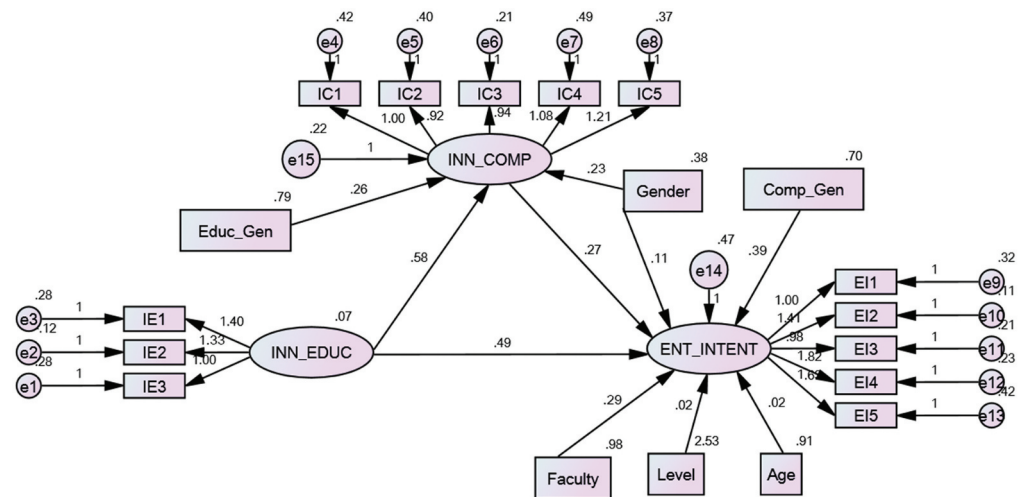


Figure 3. Interaction between innovation education and gender.

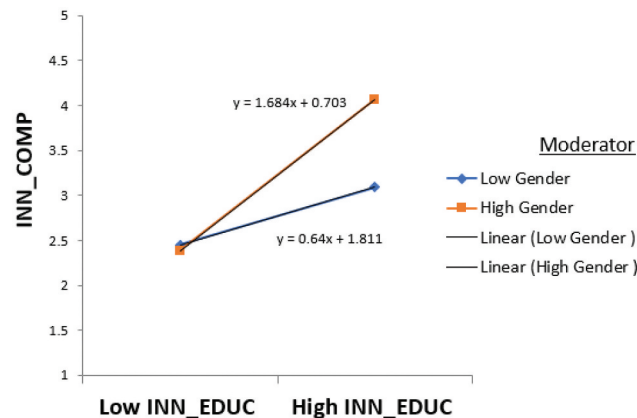
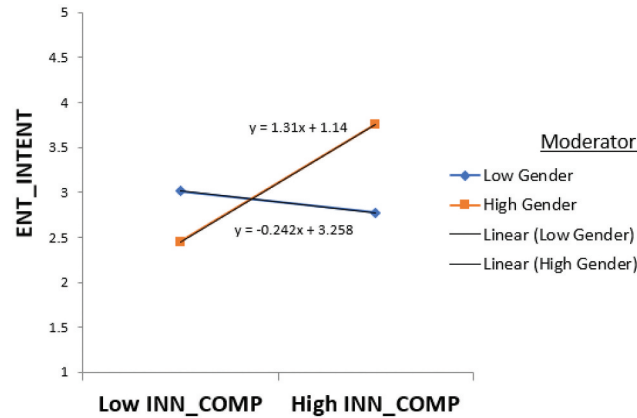


Figure 4. Interaction between innovation education and gender.



entrepreneurship. H1: “Innovation education has a direct positive effect on entrepreneurial intention among postgraduate students”, was thus supported by this study.

The effect of innovation education on innovation competence was significantly positive ($\beta = 0.581; p < 0.01$). This shows that postgraduate students with formal innovation education, were 58.1% more likely to possess innovation competence, than students with no formal innovation education. Innovation education, on the other hand, had a significant positive effect on entrepreneurial intention among postgraduate students ($\beta = 0.267; p < 0.01$). Innovation competence, therefore, enhanced postgraduate students’ entrepreneurial intention by 26.7%. With these, the indirect effect of innovation education on entrepreneurial intention was thus estimated. Results indicate that the indirect effect of innovation education on entrepreneurial intention through innovation competence was significantly positive ($\beta = 0.155$), as both the Lower and Upper BCs were positive. Innovation competence is thus said to partially mediate the relationship between innovation education and entrepreneurial intention, as innovation education also had a direct effect. H2: “Innovation competence mediates the relationship between innovation education and entrepreneurial intentions among postgraduate students”, was thus supported.

The direct effect of innovation education on innovation competence has already been established earlier. The moderating effect of gender on this relationship was thus assessed. Results from (Table 4) show that gender had a significant positive effect on innovation competence ($\beta = 0.229; p < 0.01$). This indicates that male postgraduate students were 22.9% more likely to possess innovation competence, as compared to their female counterparts. Residual centering approach to calculating the interaction term was adopted, where the residuals of the moderating and independent variables were calculated, and the results multiplied. The study identified that the interaction term between innovation education and gender (Educ_Gen) had a significant positive effect on innovation competence ($\beta = 0.261; p < 0.01$). Gender was therefore identified to have moderated the relationship between innovation education and innovation competence. That is, male postgraduate students who received innovation education possessed much higher innovation competence, compared to their female counterparts. Figure 3 demonstrates that male respondents who received innovation education had the highest score for innovation competence (orange line). Female postgraduate students who had innovation education, however, had lower score of innovation competence (blue line), compared to that of male students. H3: “Gender moderates the relationship between innovation education and innovation competence, such that the effect of innovation education on innovation competence is higher among male postgraduate students” was thus supported by the study.

The direct effect of innovation competence on entrepreneurial intention has also already been established earlier. The moderating effect of gender on this relationship was thus assessed. Results from (Table 4) show that gender had a significant positive effect on entrepreneurial intention ($\beta = 0.105; p < 0.05$). This indicates that male postgraduate students were 10.5% more likely to possess entrepreneurial intention, as compared to their female counterparts. Residual centering approach to calculating the interaction term was adopted. The study identified that the interaction term between innovation competence and gender (Comp_Gen) had a significant positive effect on entrepreneurial intention ($\beta = 0.388; p < 0.01$). Gender was therefore identified to have moderated the relationship between innovation competence and entrepreneurial intention. That is, male postgraduate students with innovation competence had much more tendency to possess entrepreneurial intention, compared to their female counterparts. Figure 4 demonstrates that male respondents with high innovation competence had the highest score for entrepreneurial intention (orange line). Female postgraduate students who had high innovation competence, however, had lower score for entrepreneurial intention (blue line), compared to that of male students. However, when innovation competence was low for both groups, female postgraduate students had higher entrepreneurial intention (blue line) compared to their male counterparts (orange line). H4: “Gender moderates the relationship between innovation competence and entrepreneurial intentions, such that the effect of innovation competence on entrepreneurial intentions is higher among male postgraduate students” was thus supported by the study.

5. Discussion of results and theoretical contribution

Innovation education was found to be positively associated with entrepreneurial intention among postgraduate students. Innovation education involves grooming, educating, nurturing, and helping students to generate and develop new business ideas for business creation (Lee & Yuan, 2018). As stated by Usman et al. (2020), innovation education has great positive influence on postgraduate students and it affects their intention to start a new business venture. Postgraduate students who go through innovation education successfully are thus able to start and properly operate a business venture.

Again, the relationship between innovation education and entrepreneurial intention among postgraduate students was found to be partially mediated by innovation competence. It was revealed in the study that innovation education enhanced innovation competence. This was in line with Keinänen and Kairisto-Mertanen (2019), who found a direct positive relationship between innovation education and innovation competence. This finding further contributes to competence-based theory (Sanchez, 2004), which stresses on using strategic and dynamic ways to accomplish successful task from insight gained by putting in every effort to meet the set target. It was further realized that innovation competence was positively associated with entrepreneurial intention among postgraduate students which was in line with studies such as Hwang et al. (2020). However, the effect of innovation education on entrepreneurial intention could also be realized through the intervention of innovation competence among postgraduate students. Usman et al. (2020) also identified similar finding such that innovation competence mediates the relationship between innovation education and entrepreneurial intention.

Moreover, despite the fact that innovation education was found to have a positive effect on innovation competence as Keinänen and Kairisto-Mertanen (2019) identified, this present study revealed that the relationship is enhanced by gender. As identified by Ferreras-Garcia et al. (2020), males are more likely to have innovation competence than females. Postgraduate male students who receive innovation education are able to acquire adequate skills and capabilities after school.

Lastly, although some previous studies such as (Schwarz et al., 2009; Joensuu et al., 2013) predicted innovation competence to have direct influence on entrepreneurial intention, this present study found that the relationship between innovation competence and entrepreneurial intention among postgraduate students is enhanced by gender. Kurczewska and Bialek (2014) identified that men are motivated enough to set up a new business venture than women do.

Although postgraduate female students also invest into starting business, postgraduate male students do it more. This study contributes to the theory of planned behavior Ajzen (1985), which stresses on putting together individual attitude, subjective norms, and perceived behavioral control to shape behavioral intentions. This is a psychological theory that links beliefs to behavior. Postgraduate male students who are believed to be innovation competent tend to become entrepreneurs than females.

6. Practical implications

Outcome of this study could be of immense relevance to the government of Ghana and policy-makers. The unemployment rate of Ghana in 2021 was 7.8% (Ghana Population and Housing Census, 2021). This figure is above the 2021 global unemployment rate of 6.4% (World Population Review, 2022). Many youths in Ghana graduate from tertiary institutions and universities but remain unemployed. The government of Ghana has therefore introduced a number of measures to reduce unemployment, such as, boost entrepreneurship through Technical and Vocational Education and Training (TVET). One strategy was the establishment of AAMUSTED, as the premier university dedicated towards entrepreneurship and TVET. Findings demonstrated that innovation education significantly influenced entrepreneurial intentions. Policy-makers must therefore seek to include innovation concept in all courses or programs run by universities, especially when teaching entrepreneurship as a subject.

Reducing unemployment through entrepreneurship will go a long way in helping reduce social vices. The general society therefore stands to benefits from the implementation of the findings from this study. Also, entrepreneurship activities will enhance the economic soundness of the country, as the performance of a country is dependent on the performance of its firms and institutions.

Also, individual students stand to benefit from the implementation of the findings of the study. Individual students will be well equipped with the necessary innovation capabilities, required for successful entrepreneurial activity after completing school. Innovation has been identified as a key driver of economic growth; therefore, these students could translate their innovation education into innovation competence, which is required for successful entrepreneurship. This could also make graduates financially independent.

7. Conclusion

The study examined the effect of innovation education on entrepreneurial intentions among postgraduate students and the role of innovation competence and gender. The study focused on Master's students of AAMUSTED. The study sampled Master's students from 10 faculties in the university to assess their behavioral intention to become an entrepreneur after school. The study used 370 students from the 10 faculties of the university as sample size based on the student population of each faculty. From the results presented, it was concluded that innovation education had a significant positive effect on entrepreneurial intention. Innovation competence was also found to partially mediate the relationship between innovation education and entrepreneurial intention. Finally, gender was found to moderate the relationship between innovation education and innovation competence, and also moderate the relationship between innovation competence.

8. Limitations and suggestions for future research

The study examined the effect of innovation education on entrepreneurial intentions among postgraduate students and the role of innovation competence and gender in Ghana. Although the results of this present study add to our knowledge, the role of innovation education in the growth of innovation competence and entrepreneurial intention among postgraduate male and female students, this study still has some limitations. First, this study looks at entrepreneurial intention and thus, we are able to conclude on which gender group is more likely to become entrepreneurs after completion of the postgraduate degree. However, this study only assessed entrepreneurial intention, and not the actual entrepreneurial activity. Also, the study focuses only on postgraduate students without involving secondary and pre-tertiary students. Future studies

should focus on how actions could be taken after entrepreneurial intention and also future studies should involve secondary and pre-tertiary students to increase the validity of the results of this study.

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